1. Business Understanding

- Business Problem & Hypotheses:

The goal question: “Who are the most likely to buy the product X?”

Additional questions: “Which business teams will consume the predictions?”, “Can we see an overview funnel of customer’s life-cycle before and after purchasing for the product?”, “Which type of customers are usually the best for this offer?”, “What is the current marketing process?”, “What is the cost of this campaign?”, “How does this campaign benefit our business?” (operations down, personalized promotion to customers?)

Target definition: “What is the prediction window?”, “Is the prediction before or after campaign execution?” → usually before the campaign, the model of after campaign is about Uplifting model.

Business Success Metrics: Conversion Rate, Marketing ROI, Lift@10, Cost per Acquisition, Revenue Lift

ML Formulation:

Ranking or probability estimation? → Usually probability-to-buy prediction
Type of problem: Binary Classification (Buy or not buy)
Should predictions be calibrated? → Yes, for the expected revenue simulation (offline evaluation)

ML Metrics:

Because overall conversion rates of a propensity-to-buy campaign are typically under 10%, there usually be imbalance problem of data. ML metric should includes: PR AUC, Precision, Recall, F1, Recall@K, Gain@K

Data Understanding:

What data sources exist? Which variables are available before campaign execution? Which are collected afterward? Which may leak information? (This is due to supervised learning and data leakage prevention)
Usually in Propensity-to-buy model, we have some main types of a customer’s information: behavioral data (page views, category browsing, previous campaign engagement, cart activity), Transactional & Operational Data (past products bought, order frequency, days since the last purchase, account age, renewal proximity, history of returns or sensitivity to promotional offers), Demographic data (profile data, account status)

Additional Data Requirements:

This is based on DS team’s domain knowledge, or through interactive discussion with marketing team to define: What useful information is missing, compare to the traditional methods of ranking?
Examples

- Customer transaction history
- Credit card spending
- Mobile app activity
- Web browsing behavior
- Previous product ownership
- Customer lifetime value

Can we access to these data? Or just in limited with the provided samples?

ML Feasibility Assessment

Can the business act on predictions? Are enough historical campaigns available? Can we solve problem without ML model?

2. Model Development & Experimentation

Dataset (How many customers? How many campaigns? Is the target balanced?)
Feature Understanding

Numerical vs categorical?

- Missing values?
- High-cardinality variables?
- Distribution shape?

Relationship analysis

Which variables correlate with conversion?

- Are there interaction effects?
- Are nonlinear relationships present?

Time Analysis

Questions

- Does seasonality exist?
- Should time-based splitting be used?
- Are campaigns independent?

Data Splitting

Questions

- Random split?
- Time split?
- Customer split?
- Cross-validation?

Data Preprocessing

Questions

Missing values

- Why are values missing?
- Missing completely or systematically?

Encoding

- One-hot?
- Target encoding?
- Frequency encoding?

Scaling

- Which algorithms require scaling?

Outliers

- Should extreme values remain?

Feature consistency

- Any duplicated customers?
- Invalid ages?
- Impossible values?

Baseline Model

Questions

- What is the simplest benchmark?
- Can business already outperform random selection?
- Does ML add value?

Examples

- Random
- Majority class
- Logistic Regression
- kNN
- Simple Decision Tree

Feature Engineering

Questions

Can we derive

- Customer tenure?
- Contact frequency?
- Days since previous campaign?
- Previous success rate?
- Interaction features?
- Behavioral aggregates?

Feature Selection

Questions

- Which features truly matter?
- Any redundant variables?
- Correlated features?
- Leakage variables?
- Stable importance across folds?

Methods

- SHAP
- Mutual Information
- Recursive Feature Elimination
- Permutation Importance

Model Training & Tracking

Questions

Which algorithms should be compared?

Examples

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM
- CatBoost

Track

- Hyperparameters
- Dataset version
- Feature version
- Metrics
- Training time
- Model artifact

Validation & Tuning

Questions

- Which CV strategy?
- Does the model well calibrated? If not, we should calibrate the model using some techniques like…
- Which metric or group of metrics should be more focused during tuning?

Final Test Evaluation

Questions

- Does test performance meet expectations?
- Is Lift acceptable?
- How stable across customer segments?

- Fairness issues?
- Explainability?

Business evaluation

Top 20% customers produce what percentage of conversions?

Offline evaluation with simulated campaigns>

3. Deployment & Operations

ML Pipeline Construction

Questions

- Batch or real-time?
- Feature pipeline reusable?
- Automated retraining?
- Pipeline monitoring?

Feature Store Integration

Questions

- Online features?
- Offline features?
- Feature consistency?
- Version control?

CI/CD Pipeline

Questions

- Automated testing?
- Model registry?
- Approval workflow?
- Rollback triggers?

System Integration

Questions

- CRM integration?
- Marketing platform integration?
- API or batch output?
- Prediction frequency?

Shadow Deployment

Questions

- Compare with current rules?
- Prediction latency?
- Distribution shift?
- User feedback?

Rollback Mechanism

Questions

- What metrics trigger rollback?
- Previous model availability?
- Automatic or manual rollback?

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4. Production Monitoring

Prediction Monitoring

Questions

- Prediction distribution stable?
- Confidence drift?
- Class proportion stable?

Data Drift Detection

Questions

- Feature distribution changed?
- New categories?
- Missing value increase?

Metrics

- PSI
- KL Divergence
- Wasserstein Distance

Concept Drift Detection

Questions

- Has customer behavior changed?
- Marketing strategy changed?
- Economic environment changed?

Model Performance Monitoring

Questions

- Precision decreasing?
- Lift decreasing?
- Calibration degrading?
- Business KPIs declining?

A/B Testing

Questions

- Does the model outperform current targeting?
- Statistical significance?
- Incremental conversion?
- Incremental revenue?

Experiment

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Control:
Random targeting
Treatment:
Model-driven targeting
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Retraining Strategy

Questions

- Scheduled retraining?
- Performance-triggered retraining?
- Drift-triggered retraining?
- Human approval?